

YUE QI

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EDUCATION

University of Glasgow

Feb 2024 - Present

PhD in Electronics and Electrical Engineering

Research Title: Multi-objective Optimisation of Integrated Energy Systems

University of Glasgow

Sep 2022 - Sep 2023

MSc in Electronics and Electrical Engineering **Distinction**

Core Modules: Electrical Energy Systems, Control, Energy Conversion Systems

China Jiliang University

Sep 2017 - Jun 2021

B.Eng. Electrical Engineering and Automation

Core Modules: Electric Drive Control System, Power System and its Automation, Principles of Protection Relay, Electric Power System, Modern Control Theory

GPA: 81.26/100

RESEARCH EXPERIENCE

22nd International Conference on AC and DC Power Transmission(Best Poster) 2025

Title: A New Model of Energy Hubs and Multi-Objective Optimisation for Future Net-Zero Distribution Networks

- Key Focus: Multi-energy integration and coordinated scheduling at the distribution level, aiming to enhance energy flexibility, carbon reduction, and operational efficiency.
- Methodology: Developed an advanced Energy Hub model incorporating electricity, heat, hydrogen, and storage; applied an improved Multi-Objective Grey Wolf Optimiser to solve a tri-objective dispatch problem.
- Impact: Provided a scalable optimisation framework for future net-zero community energy systems, contributing to low-carbon distribution network planning and operation.

The 19th IET Conference on Developments in Power System Protection 2025

- Title: Protection Insights from Optimal Operation of Offshore Energy Islands under Extreme Weather Conditions
- Key Focus: Coordinated operation between offshore energy islands and onshore grids, with a focus on reliability and protection strategy design under adverse weather scenarios.
- Methodology: Developed a joint optimisation model for offshore-onshore energy scheduling; evaluated Overcurrent (OC) protection schemes in the context of dynamic renewable penetration and extreme events.
- Impact: Offered practical protection insights for resilient grid operation, supporting the secure integration of offshore renewables into continental power systems.

M.Eng. Dissertation (Grade A) 2023

- Title: Optimisation of Distribution Grids through Symmetrical Semi-definite Programming and Renewable Distributed Energy Resources
- Key Focus: Integration of Renewable Distributed Energy Resources for enhanced grid stability and minimisation of power loss.
- Methodology: Employed Symmetrical SDP algorithm and sensitivity analyses on the IEEE 123-node distribution grid model.
- Impact: Addressed the Optimal Power Flow challenge in three-phase unbalance and contributed to a more sustainable, low-carbon energy landscape.

Integrated System Design Project (Grade A)

2023

Organiser

- Title: Reducing Carbon Emissions from The University of Glasgow
- Objective: Carbon neutrality for the University of Glasgow by 2030 through low-carbon strategies.
- Approach: Implemented vertical greening systems and heat pumps as part of a comprehensive low-carbon strategy.
- Outcomes: Found a way for a sustainable, low-carbon campus environment.

B.Eng. Project (Distinction)

2021

- Title: Economic Evaluation of Distribution Network's Peak Clipping Benefit based on Uncertainty of Distributed Generation
- Key Focus: Economic implications of integrating distributed photovoltaic generation into distribution networks.
- Methodology: Utilised MATLAB and Simulink for simulative analyses.
- Findings: Offered economic insights into the benefits of integrating low-carbon, distributed energy sources.

EDUCATIONAL AND WORK EXPERIENCE

University of Glasgow, Scotland

2024.5 – Present

Research Assistant

- Collaboration: Working with Scottish Power Energy Networks on the practical deployment and impact assessment of Low Voltage Optimisation (LVO) devices in distribution networks.
- Research Focus:
 - Conducted targeted reviews and gap analyses of protection strategies under specific deployment scenarios, leading to a conference publication.
 - Modelled and assessed the performance of various protection devices; proposed optimisation strategies aligned with current research gaps to enhance LV network resilience and efficiency.

University of Glasgow, Scotland

2024

Demonstrator

- Projects: Served as Lab Demonstrator for ENG4104 Power Systems and Eng5029 Electrical Energy Systems.
- Technical Skills: Enhanced practical skills and teaching abilities in the field of power systems.

Parker Industrial Equipment, Shanghai China

2021

Assistant Electrical Engineer

- Projects: Involved in a paint spraying and waste gas treatment project based on RTO (Regenerative Thermal Oxidizer)
- Technical Skills: Developed industrial and PLC schematic diagrams
- Safety Protocols: Conducted site safety inspections and drafted requirements

SKILLS

Technical: MATLAB, PLC

Languages: TOEFL (99), CET-6 (578); Business English Certificate (Vantage)